Here are the questions from [PSY 3500 Chapter 10 Complex Experimental Designs - Quiz]

1. A problem with designing an experiment with only two levels of the independent variable is that
A. curvilinear relationships between variables cannot be detected.
B. this design is more susceptible to confounding factors than other designs.
C. the results cannot be generalized.
D. only one dependent variable can be used with this design.

2. The simplest factorial design is the one that has
A. two independent variables with two levels each.
B. one independent variable with three levels.
C. six conditions and three possible main effects.
D. four possible interactions and eight conditions.

3. Which of the following is the general format for describing factorial designs?
A. Number of levels of first IV × Number of levels of second IV × Number of levels of third IV
B. Number of conditions of first IV × Number of conditions of second IV
C. Number of IVs × Total number of levels
D. Number of main effects × Number of interactions

4. In a factorial design, a main effect is the effect each _____ variable has by itself.
A. independent
B. correlated
C. situational
D. dependent

5. A researcher studies the effect of room music (fast, slow) and room cleanliness (tidy, untidy) on people's dining experience at a restaurant. How many main effects are possible?
A. 2
B. 1
C. 8
D. 4

6. For the same study (effect of room music (fast, slow), room cleanliness (tidy, untidy)), how many interaction effects are possible?
A. 1
B. 2
C. 8
D. 6

7. A simple main effect analysis examines
A. mean differences at each level of the independent variable.
B. relationships between independent and dependent variables.
C. the overall effect of one independent variable.
D. the overall effect of the interaction.

8. Which of the following best describes a mixed factorial design?
A. It uses both independent groups (between-subjects) and repeated measures

(within-subjects) variables.
B. It always assigns different groups of participants to each of the conditions.
C. It either uses a between-subjects design or a within-subjects design.
D. It typically assigns the same individuals to participate in all the conditions.

9. What's the design of a study that manipulates the temperature in a room (warm or cold), difficulty of study material (easy, difficult), and amount of material (low, moderate, high)
A. 2 × 2 × 3
B. 2 × 2 × 2
C. 2 x 2
D. 2 x 3

10. In a 2 × 3 factorial design, there are _____ main effect(s) and _____ interaction effect(s) possible.
A. 2; 1
B. 3; 2
C. 2; 3
D. 2; 2

Thank You